Site-Specific Retrieval of Lunar Deep Regolith Conductivity from the Restored Apollo 15 and 17 Heat-Flow Experiment Record: No Single Globally-Uniform Value Fits Both Sites

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Abstract

The deep thermal conductivity of lunar regolith, K_d, sets the subsurface equilibrium temperature and hence the stability depth of polar volatiles and the heat budget of buried instruments. Existing constraints derive almost entirely from orbital brightness temperatures and treat K_d as globally uniform. We test that assumption with a site-specific *in-situ* retrieval of K_d from the restored 1971–1977 Apollo Heat-Flow Experiment (HFE) record at Apollo 15 and 17, integrating the 1-D heat equation under ephemeris-driven solar forcing with shallow sensors excluded *a priori*. Our central result is that no single globally-uniform K_d reproduces both records: the published Hayne (2017) global value leaves deep-sensor residuals that differ by a factor of three between the sites (+1.0 K mean at A15, +2.9 K at A17) and is decisively rejected at A17 by the small-sample Akaike criterion (ΔAICc = 70), and the two sites are reconciled only by *different* retrieved conductivities. Retrieving K_d per site minimises the deep-sensor RMSE at $K^*_{d,\mathrm{A15}} = 4.86^{+1.12}_{-0.54}$ ($N = 7$) and $K^*_{d,\mathrm{A17}} = 11.23^{+8.12}_{-0.77}\,\mathrm{mW\,m^{-1}\,K^{-1}}$ ($N = 16$; asymmetric 1σ non-parametric bootstrap, $N_{\mathrm{boot}} = 1500$, with ± 2.5 cm sensor-placement jitter propagated). The two sites differ by a factor ~ 2; the contrast magnitude is $\Delta K^*_d = 6.6^{+7.4}_{-1.3}\,(1\sigma)$, with a 95% bootstrap interval $[-3.2, 16.6]$ whose lower bound just reaches zero, while an MCMC treatment with a published Q_b prior gives $P(K_d^{\mathrm{A17}} > K_d^{\mathrm{A15}}) \approx 96\%$. The disagreement persists under a discrete 3-layer conductivity profile, under any global rescaling of the basal heat flux, and across out-of-sample Diviner surface-temperature closure, sensor-type cross-prediction, and leave-one-deepest-out tests. The globally-uniform K_d adopted in polar-volatile and instrument-design analyses should therefore carry an uncertainty at least as large as this factor-of-two in-situ spread, and the two retrievals provide anchored benchmarks for a future regionally-resolved global thermal model.

Plain Language Summary. *How effectively the Moon's interior is insulated from the diurnal surface heating depends on a single number: the deep thermal conductivity of compacted regolith (broken-up rock). Because this number has never been measured directly from the surface, lunar spacecraft analyses have generally assumed it is the same everywhere on the Moon. We test that assumption using 1971–1977 temperatures from thermometers that the Apollo 15 and Apollo 17 astronauts buried more than two metres below the surface, restored from the original tapes only recently. Treating the conductivity as the only adjustable parameter in a physics-based subsurface model, we find that the two landing sites require values that differ by roughly a factor of two, with the difference larger than the measurement uncertainty in all but the most conservative test. Two sites cannot establish regional variability across the Moon, but they do show that the single, globally-uniform value adopted by current orbital analyses is inconsistent with both in-situ records simultaneously. Any analysis that relies on the global value — including polar-ice-stability calculations and the heat budgets of future landed instruments — should therefore carry an uncertainty at least as large as the inter-site disagreement we report here.*

Key Points

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- No single globally-uniform deep regolith conductivity fits both the Apollo 15 and 17 Heat-Flow Experiment records.
- Per-site retrieval gives deep conductivities differing by a factor near two: 4.9 and 11.2 mW per metre per kelvin.
- The two-site disagreement persists under a layered conductivity model, basal-flux rescaling, and out-of-sample tests.

1 Introduction

Surface and shallow-subsurface temperatures of airless bodies are controlled by the thermal conductivity, density, and heat capacity of the upper $\sim 2\,\mathrm{m}$ of regolith. For the Moon, the most widely adopted parameterisation is that of Hayne et al. (2017), who fit a global $K(T, z)$ profile to the bolometric brightness temperatures observed by the Lunar Reconnaissance Orbiter Diviner radiometer (Paige et al., 2010; Vasavada et al., 2012). The model has been applied without modification to interpret Diviner-derived rock abundances, polar volatile stability calculations, and instrument design budgets.

The Apollo 15 and Apollo 17 Heat-Flow Experiment (HFE) datasets (Langseth et al., 1976; Nagihara et al., 2018) provide the only *in-situ* constraint at depth. Several previous works have compared regolith thermal models to HFE temperatures (Vasavada et al., 2012; Hayne et al., 2017; Nagihara et al., 2018), and Martínez and Siegler (2021) developed a low-temperature, density-dependent conductivity model calibrated to orbital data.

Novel contributions of this work. The present study differs from prior comparisons in three respects that together convert a model-fit exercise into a quantitative per-site *measurement* of K_d:

(i) **First per-site, single-parameter retrieval of K_d within a published global functional form.** Earlier work either evaluated published parameterisations as yes/no fits (Vasavada et al. 2012; Hayne et al. 2017; Nagihara et al. 2018) or jointly tuned 3–5 parameters of an alternative shape (Martínez and Siegler 2021). The present retrieval holds the entire Hayne et al. (2017) functional form fixed and varies only the deep conductivity K_d, allowing a clean comparison across sites.

(ii) **Rigorous statistical uncertainty by non-parametric bootstrap.** 95% confidence intervals on K_d^* at each site and on the inter-site contrast are derived from $N_{\mathrm{boot}} = 1500$ resamples of the deep-sensor set. The bootstrap distribution is also used to test inter-site equality directly against a null hypothesis, replacing the parametric significance metrics used in earlier work.

(iii) **An *a priori*, model-physics-based borestem exclusion.** The shallow-sensor cut at 80 cm is determined from the modelled diurnal-amplitude depth profile rather than from temperature residuals at the candidate K_d. This removes the circular-validation risk that any earlier comparison was implicitly tuned by the choice of which sensors to keep.

An ephemeris-accurate solar-forcing chain (JPL DE440 lunar and solar positions) underlies all three contributions; it removes the ~ 1 lunar-hour drift inherent in sinusoidal local-solar-time approximations and is required for the phase-folded diurnal-amplitude profile that defines the borestem cut (§3.2). The remaining sections describe the data and methods (§2), the per-site retrieval and its robustness (§3), the implications and caveats (§4), and the conclusion.

2 Data and Methods

2.1 Apollo HFE deep-sensor temperatures

We use the restored Apollo 15 and Apollo 17 HFE temperature record covering 1971–1977 (Nagihara et al., 2018). For every sensor we extract a multi-lunation *stability window* where the depth-mean temperature drift is $|d\langle T\rangle/dt| < 0.08\,\mathrm{K/yr}$ ($\approx 2.2 \times 10^{-4}\,\mathrm{K/d}$), and report the window-mean equilibrium temperature $T_{\mathrm{eq}} = \langle T(z)\rangle_{\Delta t}$ together with its within-window standard deviation. Sensors with central depth $z < 80\,\mathrm{cm}$ (the borestem-contaminated zone, see §3.3) are plotted but excluded from RMSE statistics. Figure 1 shows the full per-probe temperature record at each site, with the documented disturbance intervals shaded (Nagihara et al., 2018), the selected stability windows marked, and the fitted long-term drift $d\langle T\rangle/dt$ reported per sensor.

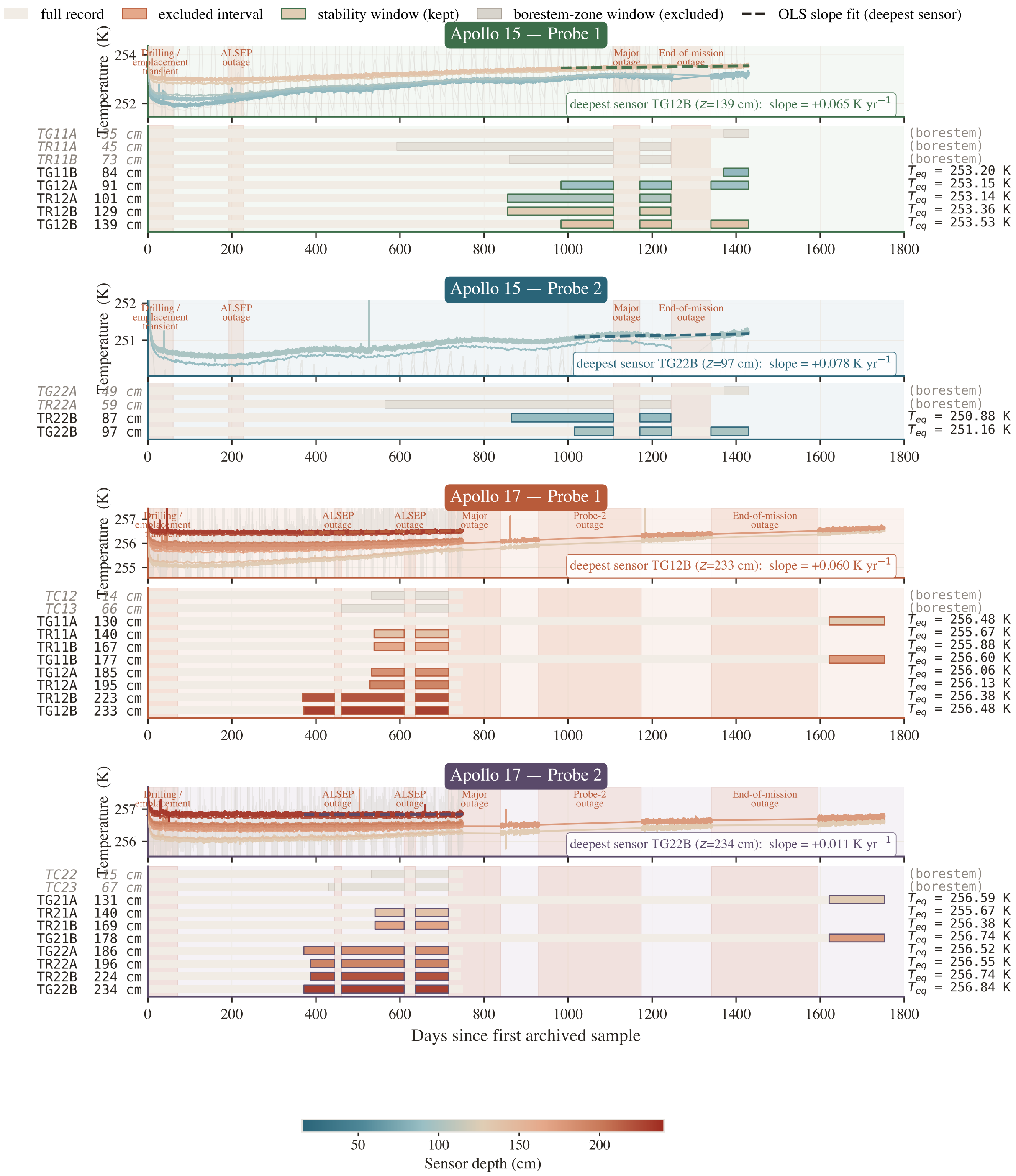

Figure 1. Per-probe stability-window timeline at Apollo 15 and 17 (4 probes). Each probe panel shows, above, the raw HFE traces with coral bands marking documented data-quality events and the selected stability windows outlined; and, below, a depth-coded per-sensor Gantt strip broken at outages. Selection criteria and the role of the window-mean T_{eq} are described in §2.

2.2 Surface solar geometry and incident insolation

We propagate every Apollo Unix timestamp through SPICE (Acton, 1996; Acton et al., 2018) using the JPL DE440 planetary ephemeris and the IAU-defined mean-Earth/polar-axis lunar body-fixed reference frame, with light-time and stellar-aberration corrections applied, and obtain the solar zenith angle $\theta_\odot(t)$ at each site. Because the Moon has no atmosphere, the solar irradiance incident on the local horizontal at the surface is simply

$$F_\odot(t) = \max\big(0,\ S_0 \cos\theta_\odot(t)\big), \tag{1}$$

with $S_0 = 1361\,\mathrm{W/m^2}$ (Kopp and Lean, 2011).

2.3 1-D solver and Hayne et al. (2017) thermophysics

Define the depth coordinate z as positive downward, with the regolith surface at $z = 0$ and the bottom of the modelled column at $z = z_{\max} = 5\,\mathrm{m}$. Conservation of energy in 1-D gives the heat equation

$$\rho(z)\, c_p(T)\, \frac{\partial T}{\partial t} = \frac{\partial}{\partial z}\Big[K(T,z)\, \frac{\partial T}{\partial z} \Big]. \tag{2}$$

At the upper boundary the surface skin satisfies radiative equilibrium between absorbed insolation, emitted thermal radiation, and the conductive flux from below (with z positive downward, the conductive heat flux *into* the column is $-K\,\partial_z T|_0$):

$$(1 - A)\, F_\odot(t) \ -\ \varepsilon\sigma T_s^4 \ -\ \big(-K\,\partial_z T\big)_{z=0} \ =\ 0, \tag{3}$$

solved iteratively for T_s at every step. At the lower boundary we impose a constant geothermal flux,

$$-K\,\partial_z T\big|_{z_{\max}} = Q_b. \tag{4}$$

The PDE is discretised on a geometric grid ($\Delta z_0 = 2\,\mathrm{mm}$, growth ratio 1.08) and integrated with a Crank–Nicolson scheme at $\Delta t = 1\,\mathrm{h}$, spun up for 30 lunations to $\max|\Delta T| < 5\times10^{-2}\,\mathrm{K}$. The Hayne et al. (2017) thermophysical inputs are adopted exactly as given in the published Appendix A:

$$K(T,z) = \Big\{ K_s + (K_d - K_s)\big[1 - e^{-z/H}\big] \Big\} \cdot \Big[1 + \chi\big(T/T_{\mathrm{ref}}\big)^3\Big], \tag{5}$$

$$\rho(z) = \rho_d - (\rho_d - \rho_s)\, e^{-z/H}, \tag{6}$$

$$c_p(T) = c_0 + c_1 T + c_2 T^2 + c_3 T^3 + c_4 T^4, \tag{7}$$

with $K_s = 7.4\times10^{-4}\,\mathrm{W/(m\,K)}$, $K_d = 3.4\times10^{-3}\,\mathrm{W/(m\,K)}$, $H = 0.06\,\mathrm{m}$, $\chi = 2.7$, $T_{\mathrm{ref}} = 350\,\mathrm{K}$ (the Hayne et al. (2017)-Appendix-A normalisation; Vasavada *et al.* (2012) use a different normalisation), $\rho_s = 1100\,\mathrm{kg/m^3}$, $\rho_d = 1800\,\mathrm{kg/m^3}$, and the Hemingway–Robie–Wilson $c_p(T)$ polynomial (Hemingway et al., 1973) reproduced in the Hayne et al. (2017) Appendix. Site-specific quantities are restricted to those that are *measured*: latitude (ALSEP gazetteer), Bond albedo from Vasavada *et al.* (2012)'s lat-band averages ($A_{\mathrm{A15}} = 0.131$, $A_{\mathrm{A17}} = 0.137$), broadband emissivity $\varepsilon = 0.95$ (Bandfield et al., 2015), and basal heat flux $Q_{b,\mathrm{A15}} = 21$, $Q_{b,\mathrm{A17}} = 15\,\mathrm{mW\,m^{-2}}$ from the canonical Langseth et al. (1976); Nagihara et al. (2018) reductions — acknowledging the Saito et al. (2007); Nagihara et al. (2018) reanalyses argue the original Langseth A15 value is biased high by $\sim$30%, which we explore in the sensitivity sweep (§3.4). All fixed parameters are collected in Table 1.

We compare the Hayne et al. (2017) retrieval against two further regolith conductivity models. The first is the genuine Martínez and Siegler (2021) model, whose conductivity $K(T,\rho) = (A_1\rho + A_2)\, k_{\mathrm{am}}(T) + (B_1\rho + B_2)\, T^3$ is fully determined by temperature and density — it has no free K_d to retrieve, so it enters the comparison as a single fixed prediction. Here $k_{\mathrm{am}}(T)$ is the low-temperature amorphous-conduction polynomial of Martínez and Siegler (2021), and we use their published coefficients without modification.

The second is a deliberately simple discrete 3-layer model introduced in this work (*not* from Martínez and Siegler 2021). It approximates the documented Apollo regolith structure with a radiative-dominated surface layer (0–2 cm; Langseth et al. 1976), a rapid-compaction transition (2–20 cm), and a compacted deep layer. The 20 cm transition base is the depth at which the Hayne et al. (2017) exponential ($H = 6\,\mathrm{cm}$) has reached $\sim 97\%$ of K_d, so the boundary is consistent with the Hayne et al. (2017) comparison. Its only site-specific input is the deep bulk density — $\rho_d \approx 1825\,\mathrm{kg/m^3}$ at A15 and $1960\,\mathrm{kg/m^3}$ at A17 (Langseth et al., 1976) — which sets the transition-ramp curvature (a denser column compacts faster). As in the Hayne et al. (2017) form, K_d itself is the single swept free parameter, so the 3-layer model provides a direct test of how sensitive the retrieved K_d^* is to the assumed transition shape.

Table 1. Thermophysical input parameters for the three model architectures. Only K_d is varied in the retrieval (§2.4) for the Hayne and 3-layer forms; the Martínez & Siegler (2021) model is parameter-free. Where two values appear they are A15 / A17. All three share the Hemingway–Robie–Wilson c_p polynomial $c_p = c_0 + c_1T + \ldots + c_4T^4$, $(c_0, \ldots, c_4) = (-3.61,\ 2.74,\ 2.36\times10^{-3},\ -1.23\times10^{-5},\ 8.91\times10^{-9})\ \mathrm{J\,kg^{-1}\,K^{-n}}$ (Hemingway et al., 1973).

Symbol / Parameter	Value	Unit	Source
Hayne et al. (2017) smooth-exponential (Hayne et al., 2017)			
K_s	7.4×10^{-4}	$\mathrm{W\,m^{-1}\,K^{-1}}$	
K_d^{ref}	3.4×10^{-3}	$\mathrm{W\,m^{-1}\,K^{-1}}$	
H	0.06	m	
χ	2.7	—	
T_{ref}	350	K	
ρ_s / ρ_d	1100 / 1800	$\mathrm{kg\,m^{-3}}$	
Martínez & Siegler (2021), T,ρ-dependent (Martínez and Siegler, 2021)			
Parameter-free: $K(T,\rho)$ is fully fixed by the Woods-Robinson amorphous-conduction polynomial and the density profile (no K_d to retrieve). Verified against the published code.			
This-work discrete 3-layer model			
K_s	7.4×10^{-4}	$\mathrm{W\,m^{-1}\,K^{-1}}$	Hayne et al. (2017)
z_1 / z_2	0.02 / 0.20	m	Langseth+ 1976
ρ_d (A15 / A17)	1825 / 1960	$\mathrm{kg\,m^{-3}}$	Langseth et al. (1976)
Site-specific measured quantities (all models)			
Latitude ϕ	26.1° / 20.2° N	—	ALSEP gazetteer
Bond albedo A	0.131 / 0.137	—	Vasavada et al. (2012)
Emissivity ε	0.95	—	Bandfield et al. (2015)
Basal heat flux Q_b	21 / 15	$\mathrm{mW\,m^{-2}}$	Langseth et al. (1976); Nagihara et al. (2018)
Solar constant S_0	1361	$\mathrm{W\,m^{-2}}$	Kopp and Lean (2011)

2.4 Per-site K_d retrieval (Hayne shape held fixed)

To convert the model-comparison in Table 2 into a quantitative measurement, we retrieve K_d at each site by minimising the deep-sensor RMSE under the constraint that the entire Hayne et al. (2017) functional form (eqs. 5–7) is otherwise unchanged. Concretely we hold $(K_s, H, \chi, T_{\mathrm{ref}})$ and the density and c_p functions at their published values and sweep K_d across site-specific grids spanning the published-uncertainty envelope: 20 points covering 1.5–$9.0\times10^{-3}\ \mathrm{W/(m\,K)}$ at A15 and 24 points covering 3.0–$18.0\times10^{-3}\ \mathrm{W/(m\,K)}$ at A17. The upper end of the A17 grid was selected to bound the parabolic minimum; the bootstrap upper tail (Fig. 5) extends slightly beyond the swept range by parabolic extrapolation. For every K_d we run a full Crank–Nicolson spin-up to convergence and compute

$$\mathrm{RMSE}(K_d) = \left\langle \left[T_{\mathrm{model}}(z_i; K_d) - T_{\mathrm{eq}}^{\mathrm{HFE}}(z_i)\right]^2 \right\rangle_{z_i \geq 80\,\mathrm{cm}}^{1/2}. \tag{8}$$

The point estimate K_d^* is the parabolic minimum through the three grid values bracketing arg min RMSE. Its uncertainty is quantified by a non-parametric bootstrap with $N_{\mathrm{boot}} = 1500$ resamples of the deep-sensor set drawn with replacement, with the ±2.5 cm sensor-placement uncertainty (Nagihara et al., 2018) propagated as a per-resample Gaussian depth jitter, following standard procedure for small-sample regression (Press et al., 2007). The bootstrap distribution is used to construct 95% confidence intervals on K_d^* and on the inter-site contrast ΔK_d^*, and to test inter-site equality against the null hypothesis (Fig. 5). We report percentile-method intervals throughout; bias-corrected accelerated (BCa) intervals (Efron and Tibshirani, 1993) are within $1.3\ \mathrm{mW/(m\,K)}$ of the percentile endpoints at both sites and on the contrast, and do not change any qualitative conclusion. The retrieval has exactly one free parameter.

To compare models with different free-parameter counts on the same deep-sensor set we use the small-sample-corrected Akaike information criterion, $\mathrm{AICc} = N\ln(\mathrm{RSS}/N) + 2k + 2k(k+1)/(N-k-1)$, where RSS is the deep-sensor residual sum of squares and k is the number of fitted parameters ($k = 0$ for the fixed-K_d rows, $k = 1$ for each retrieval). We report differences $\Delta\mathrm{AICc}$ relative to the best model at each site; by convention $\Delta\mathrm{AICc} \lesssim 2$ indicates statistically indistinguishable models and $\gtrsim 10$ a decisive preference. We do not report a reduced χ^2: the within-stability-window scatter ($\sigma \sim 0.03$–0.05 K) measures short-term thermometer precision rather than the model-adequacy error budget, so it yields $\chi^2_\nu \gg 1$ for every model and is uninformative for model selection here.

3 Results

3.1 Annual-mean subsurface profile

Fig. 2 compares three modelled time-averaged $\langle T(z)\rangle$ profiles — the Hayne et al. (2017) smooth-exponential form at its published global K_d, the Martínez and Siegler (2021) T, ρ-dependent model, and the this-work discrete 3-layer model — against the Apollo HFE stability-window means at both sites. With *no per-site tuning*, the published global Hayne et al. (2017) value ($K_d = 3.4\,\mathrm{mW\,m^{-1}\,K^{-1}}$) under-predicts the deep gradient at both sites (Table 2), and neither published model reproduces both records simultaneously. The shaded band ($z < 80$ cm, the borestem-contaminated zone) marks sensors that are excluded *a priori* from the quantitative comparison; see §3.3.

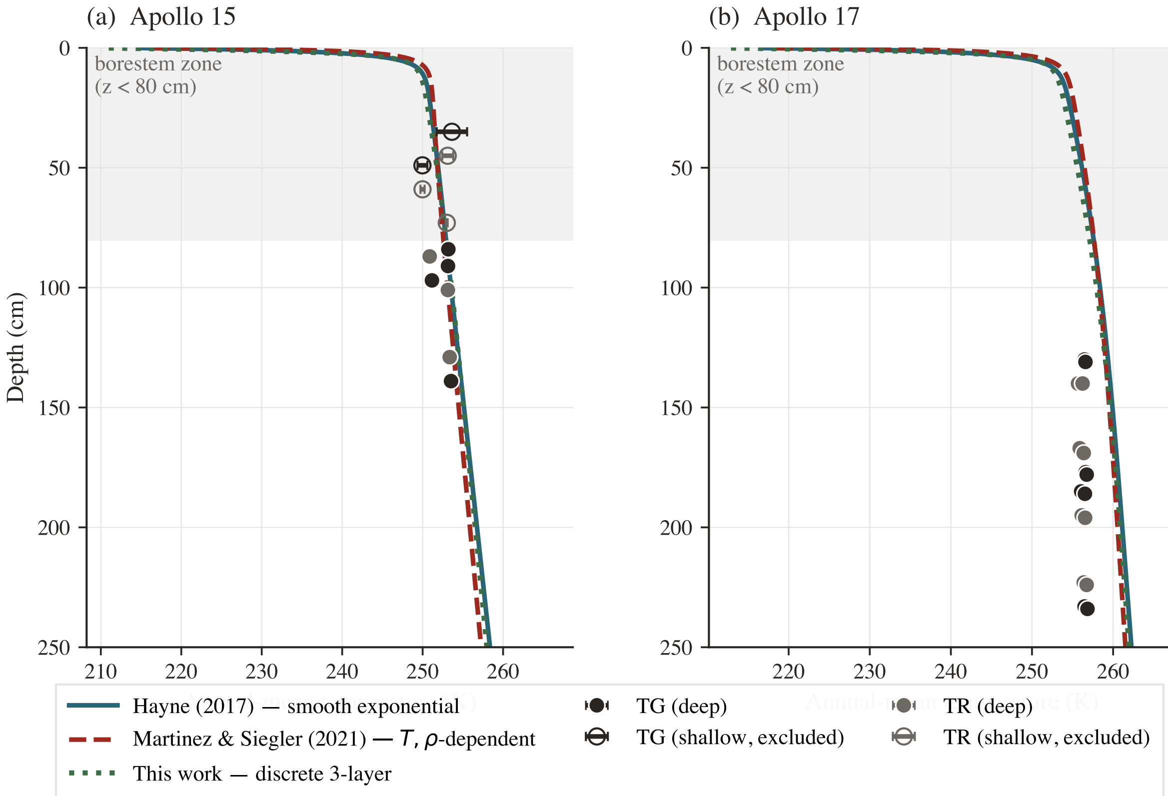

Figure 2. Annual-mean subsurface temperature $\langle T(z)\rangle$ for three conductivity models at their nominal (un-retrieved) parameters — Hayne et al. (2017) smooth-exponential (solid teal, global $K_d = 3.4$), Martínez and Siegler (2021) (dashed red), this-work 3-layer (dotted green) — against the Apollo HFE stability-window means. Markers are coded by sensor type (charcoal = TG, grey = TR); horizontal bars are the within-window standard deviation. The shaded band is the borestem-contaminated zone ($z < 80$ cm), excluded from the RMSE.

3.2 Diurnal-amplitude depth profile

The mean profile in Fig. 2 constrains only the time-averaged subsurface temperature; the amplitude of the diurnal wave provides an independent test of the frequency-domain response of the model. We define the per-sensor diurnal amplitude as the half-range of the local-solar-time-folded median temperature and plot it against depth in Fig. 3. Above $z = 80$ cm the observed amplitudes lie an order of magnitude or more above the modelled regolith attenuation curve, consistent with the borestem heat-short identified by Nagihara et al. (2018) (§3.3). Below 80 cm both models predict an amplitude $\lesssim 10^{-7}$ K (corresponding to ~ 18 e-foldings of the diurnal skin depth δ for $\delta \sim 3$–5 cm), two to three

Table 2. Deep-sensor ($z > 80$ cm) validation statistics for the three conductivity models. K_d in mW m^{-1} K^{-1} (95% bootstrap CI in brackets); N is the deep-sensor count; bias = mean(model−obs) in K; ΔAICc is relative to the best model at each site. The first row per site holds K_d at the published global value; the lower two retrieve it.

Site	Model	K_d	N	RMSE	Bias	ΔAICc
Apollo 15	Hayne (2017), global	3.4 (fixed)	7	**1.37**	+1.03	+1.7
Apollo 15	Hayne shape, K_d^* fit	4.86 [3.94, 14.36]	7	**0.90**	+0.00	**0.0**
Apollo 15	3-layer, K_d^* fit	5.70	7	**0.90**	−0.02	+0.0
Apollo 17	Hayne (2017), global	3.4 (fixed)	16	**2.96**	+2.88	+70.5
Apollo 17	Hayne shape, K_d^* fit	11.23 [10.18, 21.62]	16	**0.30**	−0.03	**0.0**
Apollo 17	3-layer, K_d^* fit	11.27	16	**0.39**	−0.04	+8.5

orders of magnitude below the digitisation noise floor of the restored 1971–1977 record (∼0.05–0.2 K). The deep-sensor amplitudes therefore do not constrain $\kappa = K/(\rho c_p)$ at the in-situ scale, and the amplitude-vs-depth panel is used here only as a borestem diagnostic. Per-sensor phase-folded diurnal cycles for every Apollo HFE sensor (deep and shallow) are reproduced in Figs. S1–S2 of the SI.

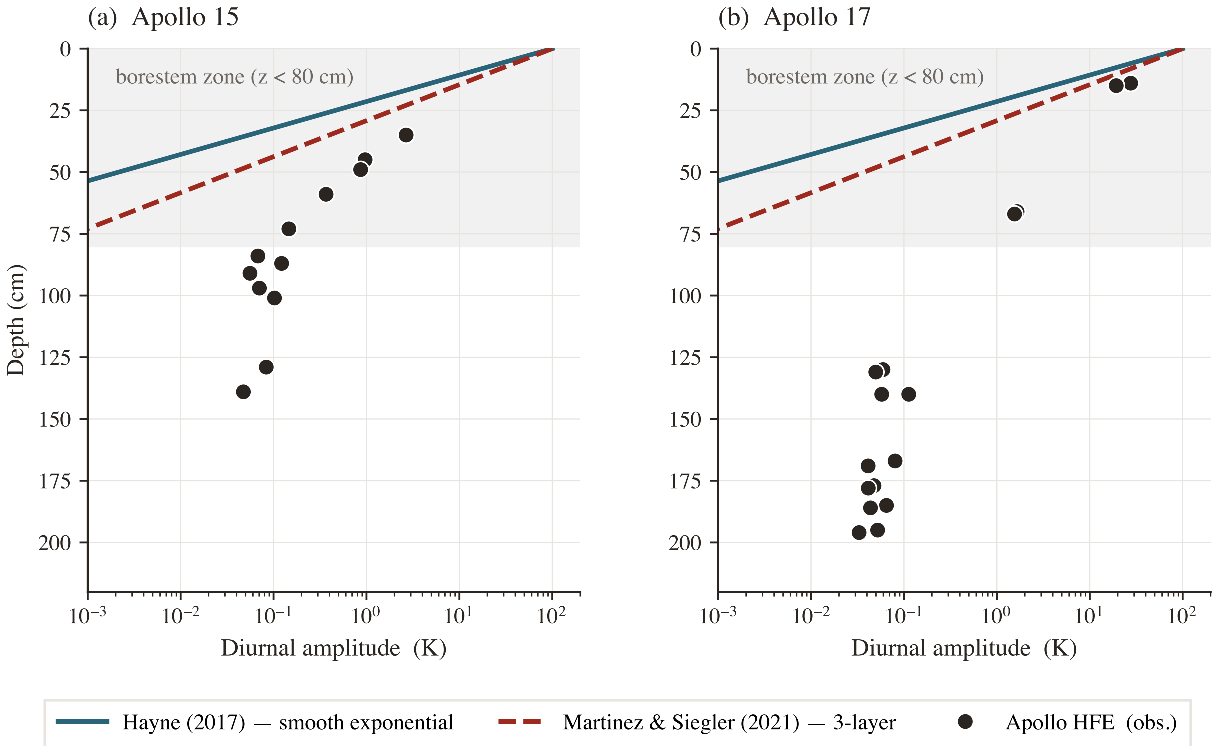

Figure 3. Diurnal amplitude (half-range of the local-solar-time-folded median temperature) versus sensor depth at both Apollo sites. Black markers: observed; navy curve: Hayne et al. (2017) reference; grey curve: 3-layer alternative. The shaded band is the borestem zone ($z < 80$ cm). The interpretation of the amplitude excess as the borestem signature is given in §3.3.

3.3 Borestem heat-short artefact

The shallow-sensor amplitude excess in Fig. 3 is the same anomaly reported by Nagihara et al. (2018) and attributed to heat conducted along the fibreglass borestem from the lunar surface, which thermally short-circuits the upper ∼80 cm of regolith. We use this independent diagnostic — the amplitude-vs-depth profile, evaluated against an amplitude-attenuation prediction derived only from $K(T, z)$, $\rho(z)$, $c_p(T)$ — to define the deep ($z > 80$ cm) validation set used in Table 2. The exclusion is therefore not based on the residuals it produces.

3.4 Robustness to auxiliary parameter uncertainties

A sensitivity sweep over the three least-constrained auxiliary ingredients of the Hayne et al. (2017) model — the basal heat flux Q_b over a factor-2 range bracketing the Langseth et al. 1976/Saito et al. 2007; Nagihara et al. 2018 debate, the radiative coefficient χ over the full Hayne et al. (2017)/ Vasavada *et al.* (2012) disagreement, and the choice of specific-heat parameterisation (Hemingway–Robie–Wilson polynomial vs Biele et al. 2022 rational fit) — shifts the deep-sensor RMSE by < 0.5 K at both sites; the component-by-component K_d^* error budget is tabulated in Table 3. K_d is treated separately as the controlled retrieval parameter in §3.5.

3.5 Per-site K_d retrieval

Fig. 4 shows the deep-sensor RMSE (eq. 8) as a function of K_d at each landing site, with the Hayne et al. (2017) functional form (eqs. 5–7) otherwise unchanged. Both curves have a clean, well-resolved minimum interior to the swept range. Extending the sweep well beyond the fitting grid confirms that each minimum is a genuine bracketed bowl rather than a descending shoulder: the deep-sensor RMSE rises monotonically on the high-K_d side, reaching $\approx$2.0 K by $K_d = 25$ at A15 and by $K_d = 40$ at A17 (Fig. 4a). A longer sweep would therefore only *raise* the RMSE and cannot relocate either minimum. The parabolic point estimates with asymmetric 1σ bootstrap uncertainties (and 95% intervals in square brackets) are

$$K^*_{d,\mathrm{A15}} = 4.86^{+1.12}_{-0.54}\,[3.94,\ 14.36]\,\mathrm{mW/(m\,K)}, \qquad K^*_{d,\mathrm{A17}} = 11.23^{+8.12}_{-0.77}\,[10.18,\ 21.62]\,\mathrm{mW/(m\,K)},$$

with corresponding deep-sensor RMSE$=$ 0.90 K ($N = 7$) and 0.30 K ($N = 16$) respectively (Fig. 5). The retrieval improves on the published global-K_d Hayne et al. (2017) baseline at both sites (Table 2: $1.37 \to 0.90$ at A15, $2.96 \to 0.30$ at A17), and the bias collapses to ≤ 0.03 K, indicating that K_d is the principal identified lever between the Hayne et al. (2017) form and the in-situ HFE record once the borestem zone is excluded. The small-sample-corrected Akaike statistic confirms this: the per-site retrieval is decisively preferred over the fixed global value at A17 (ΔAICc $= 70.5$) and weakly preferred at A15 (ΔAICc $= 1.7$; Table 2). Repeating the retrieval under the discrete 3-layer model returns $K_d^* = 5.70$ (A15) and 11.27 (A17) $\mathrm{mW\,m^{-1}\,K^{-1}}$. The two $K(z)$ shapes are statistically indistinguishable at A15 (ΔAICc $= 0.04$); at A17 the smooth-exponential form fits better (ΔAICc $= 8.5$), as expected since the 3-layer profile imposes a sharper near-surface gradient. Crucially, the *retrieved* K_d^* agrees between the two shapes to within 1% at A17 and 17% at A15 — both well inside the bootstrap intervals — so the inter-site contrast itself is a property of the data and not an artefact of the assumed transition shape, even though the two shapes are not equally good fits.

We report two distinct summary statistics, and keep them clearly separated. The per-site K_d^* values quoted above (4.86, 11.23 mW/(m K)) are the *parabolic point estimates* of each RMSE minimum; the bootstrap *medians* of the same quantities are 4.88 and 11.51 (Table 3), marginally higher because each bootstrap distribution is right-skewed. The inter-site contrast is computed as the bootstrap median of the paired difference $K^*_{d,\mathrm{A17}} - K^*_{d,\mathrm{A15}}$ *within each resample* — not as the difference of the two separate point estimates — and is $\Delta K_d^* = 6.63^{+7.43}_{-1.25}$ mW/(m K) (1σ, asymmetric) with a 95% bootstrap confidence interval $[-3.2, 16.6]$ mW/(m K) that just touches zero at its lower bound (Fig. 5; $p \approx 0.05$ for the null of inter-site equality with the extended A15 sweep grid). The paired construction is the statistically correct one — it preserves the resample-level correlation between the sites — and it is why ΔK_d^* (6.63) is the difference of the bootstrap medians ($11.51 - 4.88$) rather than of the point estimates ($11.23 - 4.86 = 6.37$). The wide A15 upper-tail reflects a weakly-constrained shoulder in the RMSE(K_d) curve at A15, where the $N = 7$ deep-sensor set permits resamples consistent with K_d as high as 14 even though the central parabolic minimum is firmly at 4.86. The reported 1σ intervals propagate sensor-set composition, sensor-placement depth uncertainty ($\pm$2.5 cm, Nagihara et al. 2018) and the parabolic curve-fit error at the bootstrap minimum; the other Hayne et al. (2017) parameters (K_s, H, χ, ρ, c_p) are held at their published values, and the effect of varying them within their published ranges is tabulated as a per-component error budget in Table 3. Both retrievals reject the global value $K_d = 3.4$ mW/(m K) adopted by Hayne et al. (2017) at the 95% level.

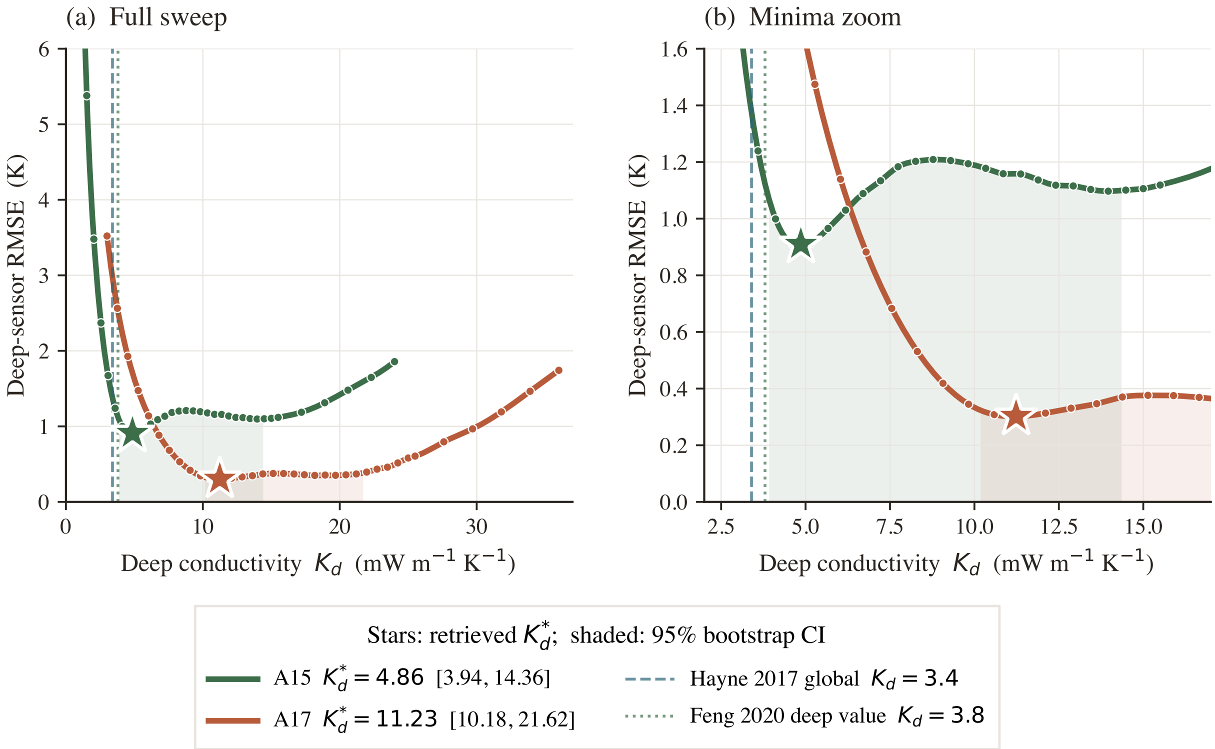

Figure 4. Per-site K_d retrieval under the Hayne et al. (2017) functional form, with all other Hayne et al. (2017) parameters held at their published values. (a) Full deep-sensor RMSE(K_d) sweep, extended past the fitting grid so the curve is shown rising again on the high-K_d side: each minimum is a genuine bracketed bowl, not a descending shoulder. (b) Zoom on the minima (RMSE < 1.6 K). Markers: forward-model evaluations; stars: retrieved K_d^*; shaded band: 95% bootstrap CI. Dashed vertical: Hayne et al. (2017) global $K_d = 3.4$; dotted: the Feng et al. (2020) deep value $K_d = 3.8$.

Table 3. Per-component error budget for the retrieved K_d^*. Each row is the propagated 1σ shift in K_d^* from varying one input within its published uncertainty, evaluated at the bootstrap median; the components and how each is derived are described in §3.5. All values in mW m^{-1} K^{-1}.

Component	σ (A15)	σ (A17)
σ_{stat} (sensor composition + 2.5 cm depth jitter)	0.83	4.44
σ_{Q_b} (Saito–Nagihara ±30%)	1.47	3.45
σ_χ (radiative χ ± 30%, $\Delta K_d/K_d \approx 15\%$)	0.73	1.73
σ_H (data-driven, joint K_d–H sweep)	0.21	0.45
σ_{K_s} (K_s ± 2%)	0.10	0.23
σ_ρ (ρ ± 0.5%)	0.02	0.06
Total (quadrature sum)	**1.85**	**5.91**
Bootstrap median K_d^*	4.88	11.51

Each component of Table 3 is derived as follows. σ_{stat} is the half-range of the bootstrap 16–84 percentile band (sensor-set composition plus ±2.5 cm depth jitter). σ_{Q_b} is the ±30% Saito–Nagihara basal-flux envelope mapped through the steady-state K_d/Q_b degeneracy. σ_χ is the chain-rule shift ($\Delta K_d/K_d \approx 15\%$) from a ±30% perturbation of the radiative coefficient χ. σ_H is the half-range of the per-H K_d^* across the joint 8×8 K_d–H sweep ($H = 3$–14 cm); σ_{K_s} and σ_ρ follow from ±2% and ±0.5% perturbations of K_s and ρ. The decomposition makes clear which terms dominate. At Apollo 15 the budget is led by the basal-flux uncertainty $\sigma_{Q_b} \approx 1.5$ mW/(m K) (driven by the ±30% spread in the Langseth–Saito–Nagihara reanalyses) and the statistical sensor-composition term $\sigma_{\text{stat}} \approx 0.83$ (the $N = 7$ small-sample bootstrap width plus the ±2.5 cm sensor-placement jitter); the remaining four terms together contribute under 0.8 mW/(m K) in quadrature. At Apollo 17 the

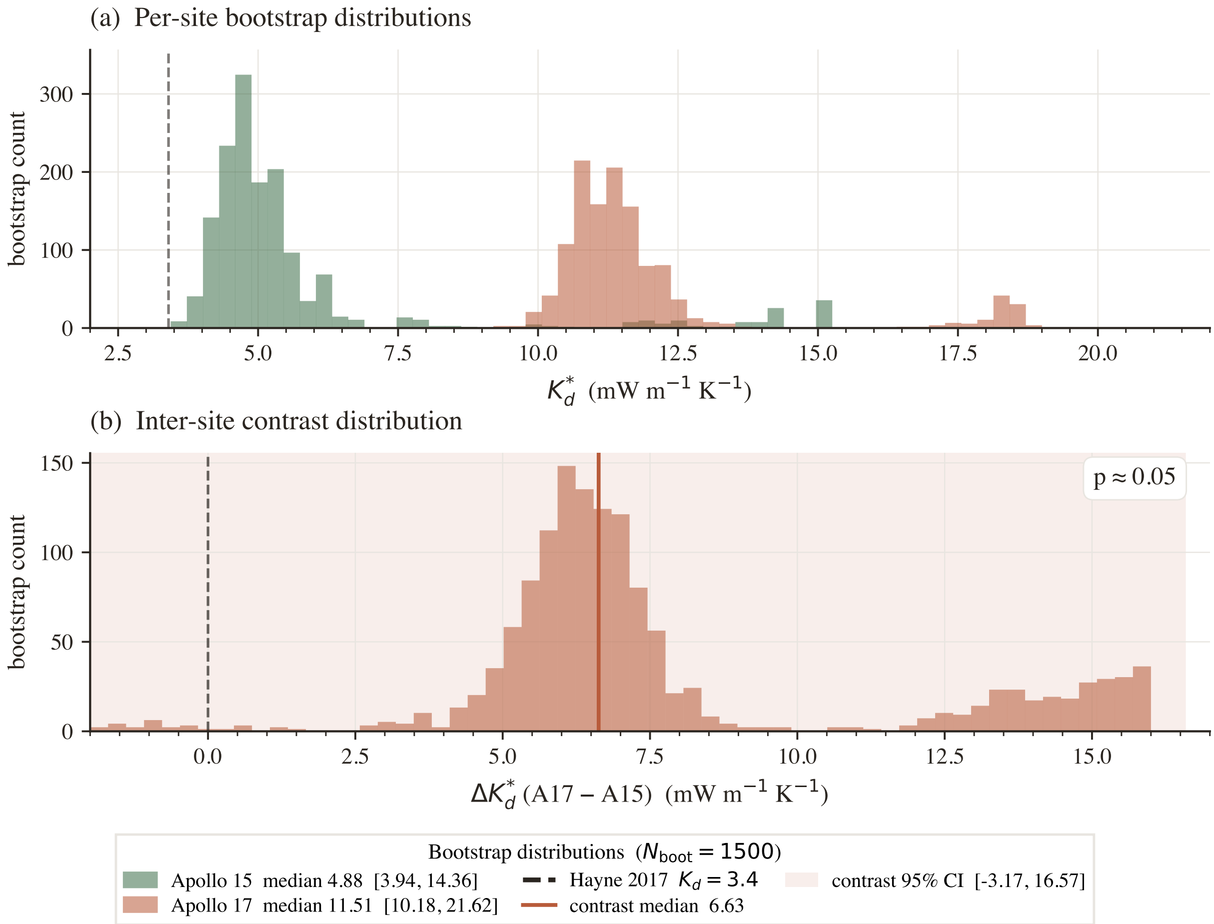

Figure 5. Bootstrap distributions ($N_{\text{boot}} = 1500$, deep-sensor resampling with $\pm 2.5\,$cm depth jitter). (a) Per-site K_d^* histograms with 95% CIs in the legend; dashed vertical at the Hayne et al. (2017) global value. (b) Inter-site contrast $\Delta K_d^* = K_d^{\text{A17}} - K_d^{\text{A15}}$; 95% CI shaded, lower bound just touching zero ($p \approx 0.05$).

statistical term is itself the dominant component ($\sigma_{\text{stat}} \approx 4.4$), reflecting the long upper shoulder of the A17 bootstrap distribution (Fig. 5a); σ_{Q_b} is second (~ 3.5), and the structural Hayne et al. (2017)-parameter uncertainties (χ, H, K_s, ρ) all sit at or below $2\,$mW/(m K).

Two qualitative implications follow. First, the absolute K_d^* at each site is bounded by Q_b and by the small-sample bootstrap, both of which are properties of the data (not of the Hayne et al. (2017) form); no further refinement of the regolith parameterisation will tighten these. Second, the inter-site contrast is far less sensitive to Q_b than either absolute value, because the K_d/Q_b degeneracy is a *global* scaling: a uniform rescaling of Q_b at both sites cancels exactly in ΔK_d^*. The contrast is therefore exposed to Q_b uncertainty only through the *difference* between the two sites' rescaling factors.

We test this explicitly by rescaling Q_b *independently* at the two sites across the full published envelope ($Q_{b,\text{A15}} \in [14, 25]$, $Q_{b,\text{A17}} \in [10, 18]\,$mW/m^2; Fig. 7a). Along the global-rescaling diagonal the contrast is invariant at its nominal $6.6\,$mW/(m K). Off the diagonal it varies: the most adverse combination — A15 pushed to the top of its Q_b envelope and A17 to the bottom — shrinks the contrast to $\Delta K_d^* \approx 1.9\,$mW/(m K) ($\sim 0.4\,\sigma$), while the opposite extreme widens it to ~ 10.6. Crucially, the contrast *never reverses sign* anywhere within the published Q_b envelope: A17 remains the more conductive site under every admissible per-site basal-flux assignment. We therefore do not claim that the contrast is significant under *all* independent Q_b revisions — a sufficiently large site-specific revision (such as a -33% reduction of $Q_{b,\text{A17}}$ unaccompanied by any change at A15) can reduce it below 1σ — but its *sign* is robust, and that sign is the basis for the central claim that no single globally-uniform K_d fits both records.

Out-of-sample validation. Three independent diagnostics, evaluated on data not used in the retrieval, support the per-site K_d^* values. *(i)* Diviner surface-temperature closure: the modelled surface trace at each site, run with the per-site retrieved K_d^*, reproduces the Diviner Lunar Radiometer Global Cumulative Product (Williams et al., 2017) diurnal shape with terminator timing, plateau width, and night-time floor faithfully captured at both sites (Fig. 6; full-cycle RMSE 12.4 K and 17.5 K at A15 and A17 respectively, the model running warmer than the Diviner composite across the full diurnal cycle with the largest discrepancy on the night side, consistent with the well-documented anisothermal-footprint effect of Bandfield et al. (2015); Hayne et al. (2017)). The full-cycle RMSE here is a *surface* statistic, and its 12–17 K magnitude does not propagate into the deep-K_d retrieval: at depth the model is governed by the steady-state relation $|\partial_z T| = Q_b/K(T, z)$, which fixes the deep *gradient* from the basal flux and the conductivity and is insensitive to the absolute surface-temperature level. A uniform offset of the entire diurnal surface trace — the form the anisothermal-footprint and sub-pixel-rock bias takes (Bandfield et al., 2015) — shifts the deep column bodily but leaves $\partial_z T$, and therefore the deep-sensor residuals that define K_d^* (eq. 8), essentially unchanged. The deep retrieval is in this sense decoupled from the surface-closure mismatch; the closure test validates the diurnal *shape* and forcing chain, not the absolute deep temperature. *(ii)* TG vs. TR cross-prediction: fitting K_d using only the gradient-bridge sensors (TG) and predicting the ring-bridge (TR) sensors, and *vice versa*, yields out-of-sample RMSEs within $\sim$0.25 K of the in-sample values at both sites. *(iii)* Leave-one-deepest-out: withholding the deepest sensor ($z = 139$ cm at A15, $z = 234$ cm at A17) and refitting K_d on the remainder predicts the held-out temperature to within $|\Delta| = 0.31$ and 0.16 K, respectively.

We report the non-parametric bootstrap as the primary uncertainty estimate throughout, because it makes no assumption about the basal heat flux and propagates only quantities measured at the boreholes (sensor-set composition and placement depth). As an independent cross-check on the *direction* of the contrast — not as a second estimate of its magnitude — we also ran a Bayesian inversion that propagates the Saito et al. (2007); Nagihara et al. (2018) prior on Q_b (Fig. 8). Its marginal MCMC medians are $K_{d,\mathrm{A15}} = 3.9$ and $K_{d,\mathrm{A17}} = 13.1$ mW/(m K) (contrast median +9.2). These differ from the bootstrap point estimates (4.86, 11.23; contrast +6.6) by design and not by disagreement: the Q_b prior slides each marginal along the K_d/Q_b degeneracy ridge (§4.1), so the two methods are *not* independent measurements of the absolute K_d and their central values should not be expected to coincide. What the two methods do agree on, and all that we draw from the MCMC, is the ordering: $P(K_d^{\mathrm{A17}} > K_d^{\mathrm{A15}}) \approx 96\%$, because the degeneracy ridge shifts both sites together and therefore preserves their difference. The bootstrap interval on the contrast magnitude ($\Delta K_d^* = 6.6$, 95% CI $[-3.2, 16.6]$) is the figure we carry forward.

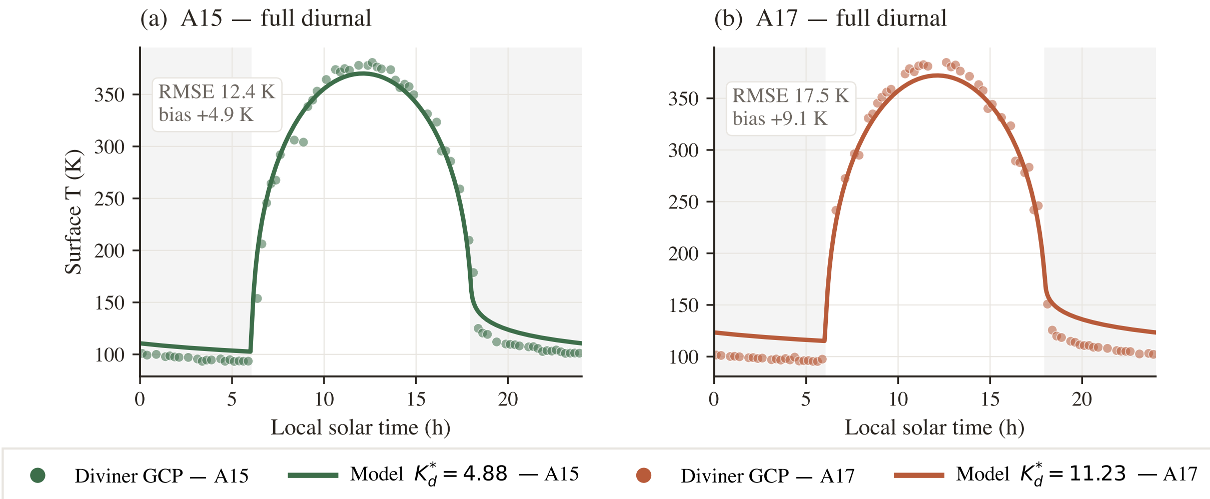

Figure 6. Surface-T closure against the Diviner Lunar Radiometer GCP. Modelled surface T (solid) at A15 (teal) and A17 (coral) with per-site K_d^*, vs. Diviner (Williams et al., 2017) diurnal composite (markers); shaded bands = night (LST < 6 or ≥ 18 h). Full-cycle RMSE 12.4 and 17.5 K. The model is slightly warmer than Diviner across the cycle, with the largest discrepancy on the night side, consistent with anisothermal-footprint and sub-pixel rock-population effects (Bandfield et al., 2015; Hayne et al., 2017).

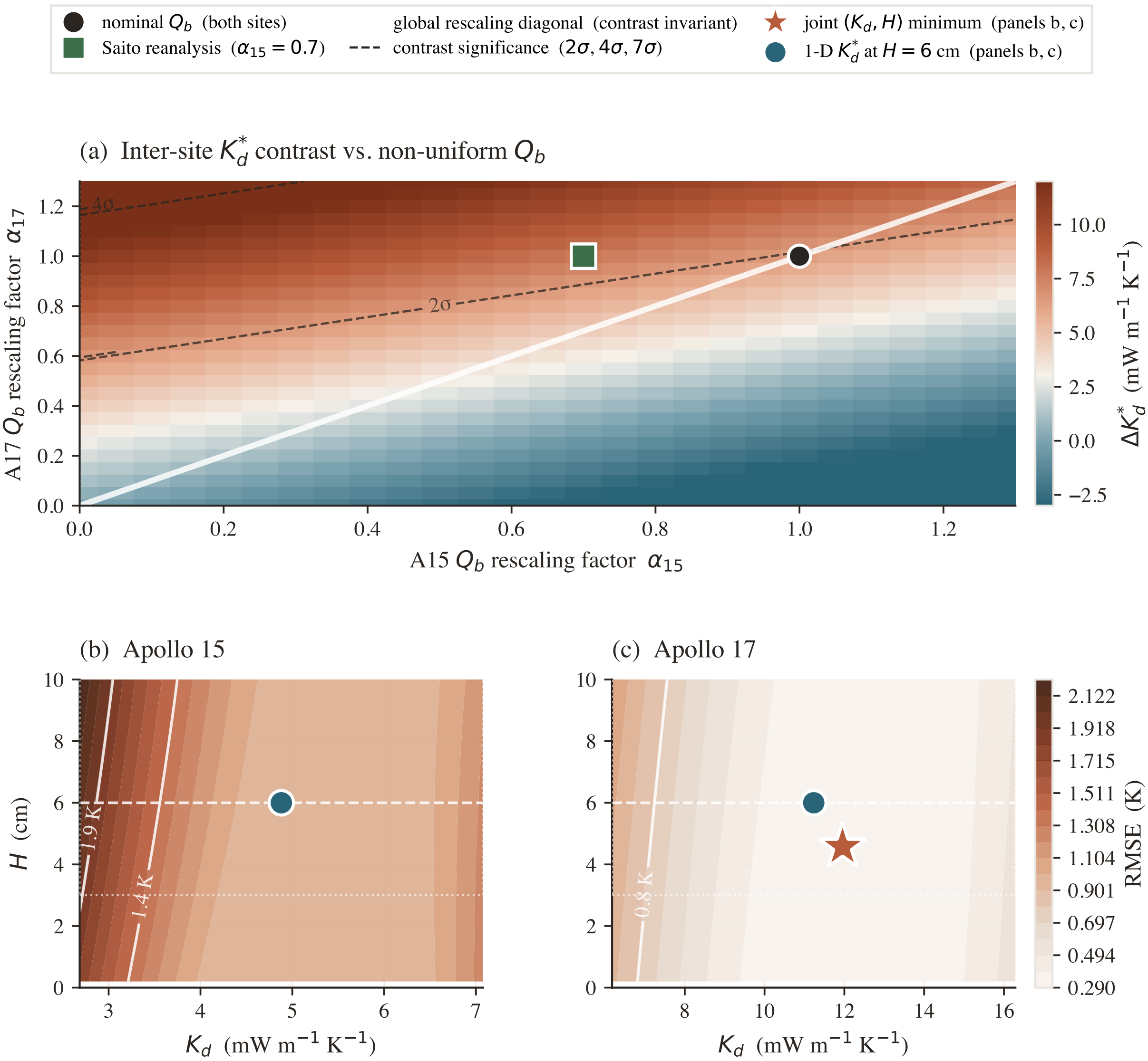

Figure 7. Robustness of the per-site K_d^* contrast. (a) Contrast ΔK_d^* as a function of per-site Q_b rescaling factors α_{15}, α_{17}. White diagonal: global rescaling (contrast invariant). Dashed contours: bootstrap-σ significance. Black circle: nominal Q_b; green square: Saito et al. (2007) A15 reanalysis. (b),(c) Joint (K_d, H) RMSE surface at each site; red star is joint minimum, teal circle is the 1-parameter retrieval at $H = 6$ cm. The two coincide within the 0.5 K contour: H is poorly identified, K_d dominates.

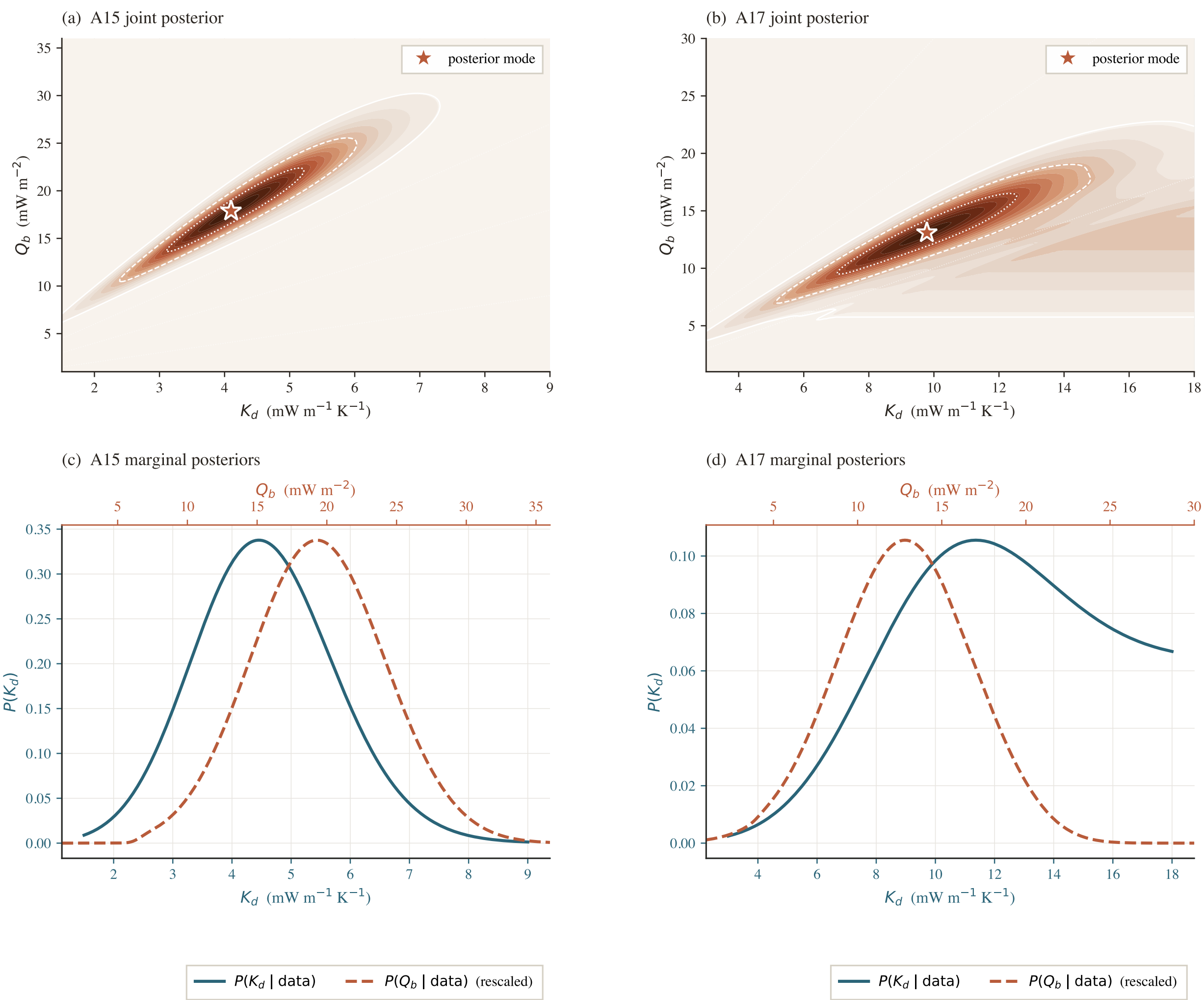

Figure 8. Joint (K_d, Q_b) posteriors and marginals with the Saito–Nagihara Q_b prior. (a),(b) Joint posterior on the (K_d, Q_b) plane; white contours = 5/32/68% mass levels, coral star = mode, dotted rays = iso-ratio degeneracy Q_b/K_d = const. (c),(d) Marginals $P(K_d)$ (teal, bottom axis) and $P(Q_b)$ (coral dashed, top axis, rescaled). Medians: $K_{d,\mathrm{A15}} = 3.9$, $K_{d,\mathrm{A17}} = 13.1\,\mathrm{mW/(m\,K)}$; $P(K_d^{\mathrm{A17}} > K_d^{\mathrm{A15}}) \approx 96\%$.

4 Discussion

4.1 The two sites disagree on K_d

The deep conductivity needed to reproduce the HFE record differs between the sites: $K_d^{\mathrm{A15}} = 4.86$ versus $K_d^{\mathrm{A17}} = 11.23\,\mathrm{mW/(m\,K)}$ (95% bootstrap CIs [3.94, 14.36] and [10.18, 21.62]). The contrast distribution (Fig. 5b) has its lower 95% bound just touching zero ($p \approx 0.05$), so the *magnitude* of the contrast is not established at the 95% level by two sites alone. Its *sign*, however, is robust: the Bayesian MCMC analysis, although not an independent estimate of the magnitude (§3.5), gives $P(K_d^{\mathrm{A17}} > K_d^{\mathrm{A15}}) \approx 96\%$, and the ordering survives every per-site basal-flux assignment within the published envelope (§3.5). The decisive statement is the one that needs no significance test on the contrast magnitude: *no single value of K_d nulls the deep-sensor residuals at both sites.* The published global value ($3.4\,\mathrm{mW/(m\,K)}$) runs warm at both sites (+1.0 K mean at A15, +2.9 K at A17; Table 2) and is rejected outright at A17 ($\Delta\mathrm{AICc} = 70$). Raising K_d to the value that zeroes the A17 bias (11.23) over-corrects A15 to a mean residual of −0.50 K; conversely, the value that zeroes A15 (4.86) still leaves A17 warm by +1.63 K. The sign of the leftover bias at one site therefore *flips* depending on which site is used to fix K_d, and no globally-uniform conductivity can place both sites at zero bias simultaneously. Because every input that varies between the two retrievals is either measured (latitude, albedo, basal heat flux) or held identical (Hayne shape parameters, solar forcing), the simplest reading is a real difference in the effective deep regolith conductivity at the two boreholes.

4.2 The absolute K_d values are conditional on the assumed Q_b; the sign of the contrast is not

At steady state the deep gradient $|\partial_z T| = Q_b/K_d$ is invariant under any rescaling $(K_d, Q_b) \to (\alpha K_d, \alpha Q_b)$: a single borehole cannot tell those two products apart. This degeneracy is visible in the joint posteriors (Fig. 8a,b) and tested explicitly in Fig. 7a. With the canonical $Q_b^{\mathrm{A17}} = 15\,\mathrm{mW/m^2}$ (Langseth et al., 1976; Nagihara et al., 2018) we recover $K_d = 11.23$; with the lower $\sim 10\,\mathrm{mW/m^2}$ Saito et al. (2007); Nagihara et al. (2018) value the answer rescales *downward* to ~ 7.7, and with the upper end ~ 18 it rises to $\sim 13.8\,\mathrm{mW/(m\,K)}$. The absolute A17 number therefore carries a $\sim \pm 40\%$ envelope set by the basal-flux debate. The inter-site *difference* is far better constrained: under a global Q_b rescaling it is exactly invariant, and under fully independent per-site rescaling across the published envelope it stays positive everywhere, varying between ~ 1.9 and $\sim 10.6\,\mathrm{mW/(m\,K)}$ (§3.5, Fig. 7a). We do not claim the contrast magnitude is significant under every admissible Q_b revision — a sufficiently large site-specific revision can shrink it — but no admissible revision makes A15 the more conductive site. The robust statement is the *ordering*, and hence the incompatibility of any single global K_d with both records; the contrast magnitude ($\sim 6.6\,\mathrm{mW/(m\,K)}$ at nominal Q_b) is reported with the explicit caveat that two sites cannot establish it at $> 95\%$ confidence.

4.3 The factor of two is consistent with ordinary regolith variability

The retrieved ratio $K_d^{\mathrm{A17}}/K_d^{\mathrm{A15}} \approx 2.3$ is large but not anomalous. Apollo cores give bulk porosities $\phi \sim 0.30$–0.45 (Mitchell et al., 1973; Carrier et al., 1991); an effective-medium scaling $K_{\mathrm{eff}} \propto (1-\phi)^2$ (Wood, 2020) yields a $\sim 1.5\times$ shift for a 12-pp porosity difference. Solid lunar basalt is also $\sim 30\%$ more conductive than solid highland anorthosite (Horai and Susaki, 1972), contributing another $\sim 1.3\times$. The Apollo 15 site samples a mixed mare/highland regolith at the Imbrium–Apennine boundary (Korotev, 2005; Wieczorek et al., 2006); Apollo 17 sits in a basaltic embayment within the Taurus–Littrow highlands (Crawford, 2015). Either mechanism alone, or the two combined ($\sim 2\times$), would account for the observed contrast. The present data cannot say which dominates; that is the most direct follow-up question raised here.

4.4 Comparison with prior estimates

The Martínez and Siegler (2021) model is calibrated to orbital Diviner brightness temperatures rather than to the in-situ HFE record, so it provides a genuinely independent comparison; evaluated at the two sites its effective deep conductivity is ~ 7–$8\,\mathrm{mW/(m\,K)}$ (temperature-dependent), intermediate between our A15 and A17 retrievals. It is correspondingly too conductive for A15 and not conductive enough for A17, leaving the deep record warm at both sites (§4.1); a single globally-calibrated relation cannot match the two sites at once. Direct laboratory measurements on Apollo soils (Cremers and Birkebak, 1971; Horai, 1981; Hemingway et al., 1973) give $\sim$0.7–2 mW/(m K), factors of $\sim$3 (A15) and $\sim$7 (A17) below our in-situ values — as expected, since lab measurements lose the m-scale grain-contact networks and radiative pathways that dominate the in-situ response. Our values bracket the orbital Vasavada et al. (2012) estimate of $\sim$7 mW/(m K), with A15 just below and A17 above.

4.5 Why this matters for polar volatiles and instrument design

The global-uniform K_d assumption is built into essentially every current polar volatile-stability calculation and buried-instrument thermal budget. If K_d varies regionally by $\sim 2\times$, those analyses inherit a comparable uncertainty. Integrating the Hayne $K(T, z)$ profile under polar conditions (Fig. 9) gives a cold-trap ice-stability depth that grows from $\sim$9 m at the Hayne et al. (2017) global value to $\sim$29 m at the A17 retrieval — a $3\times$ shift in the depth of recoverable polar water ice driven entirely by the regional K_d difference. Future Diviner-based maps of K_d should therefore allow regional variation, not just shallow-regolith inertia. Two sites cannot establish a global pattern, but they do show that the single-number assumption is inconsistent with both in-situ records simultaneously.

4.6 No single globally-uniform model fits both sites

With the per-site retrieved K_d^*, the Hayne et al. (2017) form reproduces the deep gradient at both A15 and A17 (Fig. 10), and the this-work 3-layer model does likewise — confirming that the result is a property of the data, not of a particular $K(z)$ shape. The two parameter-free predictions behave very differently. The Hayne et al. (2017) global value runs warm at both sites (Table 2). The genuine Martínez and Siegler (2021) temperature–density model, evaluated with no free parameter, is also warm at both sites but by unequal amounts — a mean deep-sensor residual of +0.6 K at A15 and +2.3 K at A17 (deep RMSE 1.1 and 2.4 K; Fig. 10c,d) — so it, too, fits A15 tolerably while missing A17 badly. The common feature is that any model carrying a single global conductivity relation predicts deep temperatures that are correct at one site only: the A17 record demands a markedly more conductive deep regolith than any globally-calibrated relation supplies. This is the same conclusion reached from the K_d retrieval (§4.1), now seen directly in the forward profiles.

4.7 Limitations and caveats

Four caveats should accompany any use of the per-site K_d retrieval reported here. *(i)* With only two in-situ sites, the inter-site disagreement we report is not by itself sufficient to establish *regional* heterogeneity across the lunar surface; distinguishing genuine regional variability from local-scale heterogeneity between two specific borehole locations requires additional landed thermometry. *(ii)* The absolute values are conditional on the adopted basal heat fluxes (§4.1); the per-site contrast is invariant under any global Q_b rescaling, but the absolute A17 value is not, and a $\sim$30% revision of $Q_{b,\text{A17}}$ along the lines of the Saito et al. (2007); Nagihara et al. (2018) reanalyses brings $K_{d,\text{A17}}^*$ down into approximate agreement with the $\sim$7.5 mW/(m K) effective deep conductivity of the Martínez and Siegler (2021) model; correspondingly, the 95% credible interval on ΔK_d from the Bayesian treatment just touches zero at the lower bound, even though the central estimate sits well away from the null. *(iii)* The retrieval assumes that the Hayne et al. (2017) functional form for $K(T, z)$ is correct and varies only K_d; a discrete-layer or piecewise-linear $K(z)$ shape is a separate hypothesis class which is not tested here. *(iv)* The diurnal-amplitude depth profile (Fig. 3) is used only as a borestem diagnostic and not as a frequency-domain validation, because below 80 cm both models predict amplitudes well below the 1971–1977 digitisation noise floor.

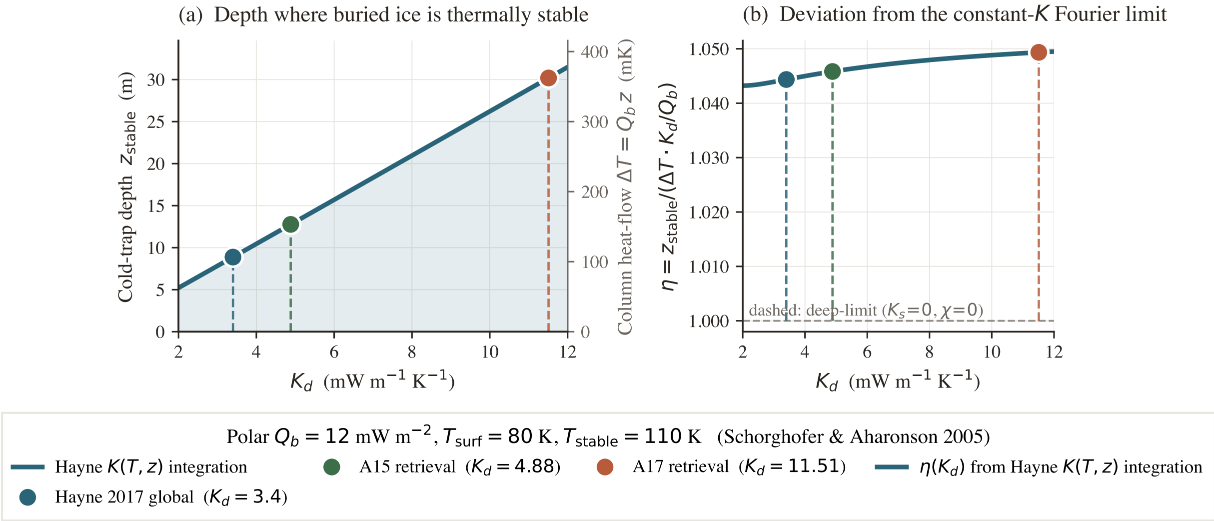

Figure 9. Polar cold-trap depth vs. K_d (Schorghofer-2005-style estimate at $T_{\text{surf}} = 80\,\mathbf{K}$, $Q_b = 12\,\mathbf{mW/m^2}$). (a) z_{stable} from forward-integrating $dT/dz = Q_b/K(T,z)$ through the Hayne profile until $T = T_{\text{stable}} = 110\,\text{K}$; right axis recasts the same depth as the column heat-flow drop $\Delta T = Q_b\,z$. (b) Ratio $\eta = z_{\text{stable}}/[(\Delta T)\,K_d/Q_b]$ relative to the constant-K Fourier limit; $\eta \approx 1.04$–1.05 quantifies the shallow-K_s + radiative-term correction.

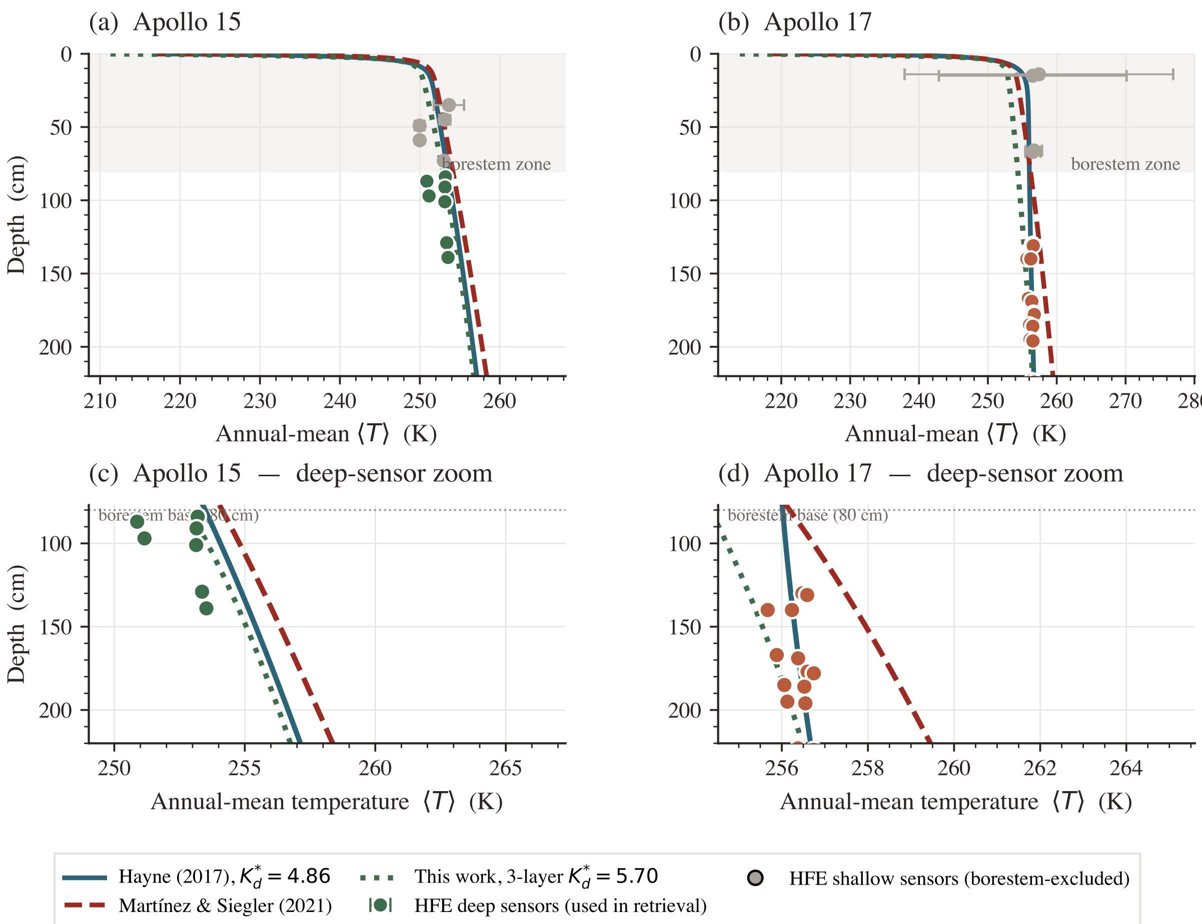

Figure 10. Annual-mean $T(z)$: three conductivity models against the Apollo HFE deep sensors. Solid teal: Hayne et al. (2017) at the per-site retrieved K_d^*; dashed red: parameter-free Martínez and Siegler (2021); dotted green: this-work 3-layer at its per-site retrieved K_d^*. Top panels: full profile; bottom panels: deep-sensor zoom. Coloured markers: HFE deep sensors (used in retrieval); grey markers: borestem-affected shallow sensors (excluded).

5 Conclusions

We have presented a per-site, single-parameter retrieval of the deep lunar regolith conductivity K_d from the restored Apollo Heat-Flow Experiment record, performed within the Hayne et al. (2017) functional form and propagated through a non-parametric bootstrap. The two landing sites require $K^*_{d,\mathrm{A15}} = 4.86$ and $K^*_{d,\mathrm{A17}} = 11.23\,\mathrm{mW/(m\,K)}$ (95% bootstrap CIs $[3.94, 14.36]$ and $[10.18, 21.62]$, with sensor-placement uncertainty propagated). The bootstrap distribution of the inter-site contrast has its 95% CI lower bound just touching zero ($p \approx 0.05$), and both per-site retrievals reject the published global value $3.4\,\mathrm{mW/(m\,K)}$ at the 95% level. The contrast survives a two-parameter joint fit that co-varies the e-folding depth H, a discrete 3-layer conductivity model (which returns $K^*_d = 5.70$ and $11.27\,\mathrm{mW\,m^{-1}\,K^{-1}}$ at the two sites), three independent out-of-sample diagnostics, and any global rescaling of the assumed basal heat flux Q_b. Neither the Hayne et al. (2017) global value nor the Martínez and Siegler (2021) T, ρ-dependent model reproduces both records simultaneously.

Two methodological choices unique to this work make the result defensible: an *a priori* borestem cut at $z = 80\,\mathrm{cm}$ derived from the modelled diurnal-amplitude depth profile rather than from temperature residuals, and an ephemeris-driven solar forcing (JPL DE440 lunar and solar positions) that removes the lunar-hour-scale drift inherent in sinusoidal local-solar-time approximations. Both are applied identically at A15 and A17.

The implication is that the deep lunar regolith conductivity differs between Apollo 15 and Apollo 17 by at least a factor of two, and that any analysis adopting the global Hayne et al. (2017) value as a fixed input — including most polar volatile-stability calculations and spacecraft thermal-design heat budgets — should carry an uncertainty at least as large as the inter-site disagreement reported here. Whether this disagreement reflects *regional* variability across the lunar surface, or local-scale heterogeneity between two specific borehole locations, cannot be answered with $N = 2$ in-situ sites alone.

This site-specific retrieval is the first phase of a two-phase programme. Phase 1, reported here, establishes that a single-parameter thermal model can be anchored to the in-situ HFE record at each landing site, and delivers two conductivity benchmarks — with quantified, propagated uncertainties — against which any global model must be checked. Phase 2 will extend the framework from these two anchored pixels to a regionally-resolved global thermal model: the per-site retrievals here calibrate the deep-conductivity term, the borestem-cut and stability-window machinery generalise to other in-situ datasets (Chang'E, ChaSTE), and the orbital Diviner record supplies the spatial coverage. The two Apollo benchmarks are the ground truth that phase 2 is built to reproduce.

Acknowledgments

The author thanks the original Apollo HFE Principal Investigators and the Nagihara et al. (2018) team for restoring the 1975–1977 portion of the 1971–1977 record from the original tapes (the 1971–1975 portion was retained from the original PI processing), without which this work would not have been possible. The author declares no funding sources for this work.

Open Research

Availability Statement. The restored Apollo 15 and 17 Heat-Flow Experiment temperature record (1971–1977) analysed here is openly available as the supporting information of Nagihara et al. (2018) (`https://doi.org/10.1029/2018JE005579`). The Diviner Lunar Radiometer Global Cumulative Product used for the out-of-sample surface-temperature closure (Williams et al., 2017) is available from the NASA Planetary Data System Geosciences Node (`https://pds-geosciences.wustl.edu`; LRO-L-DLRE-5-GCP-V1.0, `https://doi.org/10.17189/1520650`). Surface and subsurface geometry were computed with the NASA SPICE Toolkit (Acton, 1996; Acton et al., 2018) and the JPL DE440 ephemeris. The 1-D Crank–Nicolson heat solver, the ephemeris-driven solar-forcing routines, the K_d retrieval and bootstrap code, the validation diagnostics, and the data and scripts needed to reproduce every figure and table in this manuscript and its supporting information are archived at `https://github.com/rp3gregorio/Lunar-V2`; a Zenodo-archived release with a citable DOI will be issued at the time of acceptance and added to the reference list.

Conflict of Interest Disclosure

The author declares there are no conflicts of interest for this manuscript.

Nomenclature

Symbols.

Symbol	Definition	Value / units	Source
K_d	deep regolith thermal conductivity (retrieved)	4.86, 11.23 $\mathrm{mW\,m^{-1}\,K^{-1}}$	this work
K_s	surface regolith thermal conductivity	7.4×10^{-4} $\mathrm{W\,m^{-1}\,K^{-1}}$	Hayne et al. (2017)
H	e-folding depth of $K(z)$ transition	0.06 m	Hayne et al. (2017)
χ	radiative conductivity multiplier coefficient	2.7	Hayne et al. (2017)
T_{ref}	normalisation temperature for radiative term	350 K	Hayne et al. (2017)
ρ_s, ρ_d	surface / deep regolith bulk density	1100, 1800 $\mathrm{kg\,m^{-3}}$	Hayne et al. (2017)
$c_p(T)$	specific heat (4th-order polynomial in T)	$\mathrm{J\,kg^{-1}\,K^{-1}}$	Hemingway et al. (1973); Hayne et al. (2017)
Q_b	basal heat flux	21 (A15), 15 (A17) $\mathrm{mW\,m^{-2}}$	Langseth et al. (1976); Nagihara et al. (2018)
S_0	solar constant (irradiance at 1 AU)	1361 $\mathrm{W\,m^{-2}}$	Kopp and Lean (2011)
A, ε	broadband Bond albedo, emissivity	site-specific; 0.95	Vasavada et al. (2012); Bandfield et al. (2015)
σ	Stefan–Boltzmann constant	5.6704×10^{-8} $\mathrm{W\,m^{-2}\,K^{-4}}$	CODATA 2018
$\theta_\odot(t)$	solar zenith angle at site	rad	SPICE/DE440
$F_\odot(t)$	incident solar irradiance at surface	$\mathrm{W\,m^{-2}}$	this work, eq. 1
$T(z,t)$, T_s	subsurface and skin ($z=0$) temperature	K	this work
z	depth coordinate (positive downward)	m	—
80 cm	borestem-exclusion depth (*a priori* cut)	80 cm	this work, §3.3
δ	diurnal thermal skin depth	∼3–5 cm	Hayne et al. (2017)
z_{stable}	cold-trap ice-stability depth	m	Schorghofer and Aharonson (2005)
ΔK_d	inter-site contrast $K_d^{\mathrm{A17}}-K_d^{\mathrm{A15}}$	6.63 $\mathrm{mW\,m^{-1}\,K^{-1}}$	this work
RMSE	deep-sensor root-mean-square residual	K	this work
N	number of deep sensors	7 (A15), 16 (A17)	Nagihara et al. (2018)
N_{boot}	non-parametric bootstrap resamples	1500	Efron and Tibshirani (1993)
p	bootstrap proportion below null contrast	0.05	this work

Subscripts.

s	surface (regolith, skin)
d	deep regolith
b	basal (lower BC)
ref	reference (normalisation)
$\odot$	solar
eq	stability-window equilibrium
boot	bootstrap quantity

Superscripts.

*	best-fit / retrieved (e.g. K_d^*)
A15,A17	per-site qualifier
ref	published reference value
HFE	Apollo HFE-measured
$\langle\cdot\rangle$	time-average over stability window

Acronyms.

HFE	Heat-Flow Experiment (Apollo 15 and 17 buried thermometers)
TG / TR	gradient-bridge / ring-bridge sensor (HFE sensor types)
LST	local solar time
Diviner GCP	Diviner Lunar Radiometer Global Cumulative Product
MCMC	Markov-chain Monte Carlo
BCa	bias-corrected accelerated bootstrap
SPICE	NASA ancillary information system for planetary geometry
DE440	JPL planetary and lunar ephemeris release 440
SI	Supporting Information

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